Classical Field Theory — Exercise Sheet 3

Stress, improvements and angular momentum

Hints and answers

Use $T^\mu{}_\nu = p^\mu \partial_\nu \phi - \delta^\mu_\nu L$, $\eta = \mathrm{diag}(1, -1, \dots, -1)$ and the inverse-metric Hilbert convention of Lecture 3. All quantities below are smooth. The first problem has d spatial coordinates; the remaining problems have three. Problems 1–3 use Lectures 1–3. Problem 4 is a return problem after Lecture 9; its electromagnetic component conventions are supplied in the question.

Exercise 3.1 (Work by an external coupling). On $I \times \mathbb{T}^d$ consider

$$L = \frac{1}{2}\phi_t^2 - \frac{1}{2}|\nabla\phi|^2 - \frac{\lambda(t)}{4}\phi^4,$$

where I is an open interval and λ is prescribed and smooth.

(a) Derive the field equation and energy density. For a classical solution, find the derivative of its total energy, including the term due to explicit time dependence. Decide whether its physical spatial momentum $P^i = \int_{\mathbb{T}^d} T^{0i} \mathrm{d}^d x$ remains constant.

(b) Set $I = (0, \pi/2)$, $\lambda(t) = 1/\sin^2 t$ and $\phi(t, x) = \sin t$. Verify the field equation, compute its energy and check the sign and value of the energy derivative by both methods.

Hint. Keep the explicit coordinate derivative of the density in the general stress identity. Spatial periodicity removes the flux, but does not remove work done by a prescribed time-dependent coupling.

Answer. The equation is $\phi_{tt} - \Delta\phi + \lambda\phi^3 = 0$ and $E' = \lambda' \int \phi^4 \mathrm{d}^d x/4$, while every P^i is constant. In the example, $E = (2\pi)^d(\frac{1}{2}\cos^2 t + \frac{1}{4}\sin^2 t)$ and $E' = -(2\pi)^d \sin t \cos t/2 < 0$.

Exercise 3.2 (Angular momentum and weighted decay). For the autonomous scalar density $L = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi)$ on $\mathbb{R}^{1+3}$, raise the second stress index with η. For fixed distinct spatial indices i, j, define

$$M^{\mu ij} = x^i T^{\mu j} - x^j T^{\mu i}.$$

(a) Prove its local conservation on solutions, keeping the derivatives of the explicit coordinate factors.

(b) Prove that $J^{ij}(t) = \int_{\mathbb{R}^3} M^{0ij} \mathrm{d}^3 x$ is constant if, on every compact time interval, every component of $T^{\mu\nu}$ is bounded by $C(1 + |x|)^{-4-\sigma}$ for some $C, \sigma > 0$. Explain why an unweighted integrability statement for stress alone does not automatically establish the angular charge hypotheses.

Hint. The coordinate derivatives leave $T^{ij} - T^{ji}$. Apply Lecture 2's charge theorem to the current $M^{\mu ij}$, which carries one additional power of $|x|$.

Answer. The local divergence vanishes because the raised scalar stress is symmetric. The stated decay gives $|M^{\mu ij}| \le 2C(1 + |x|)^{-3-\sigma}$, so the density and compact-time spatial flux satisfy Lecture 2's integrability hypotheses.

Exercise 3.3 (Belinfante improvement from a spin identity). Suppose a smooth tensor $T^{\mu\nu}$ on $\mathbb{R}^{1+3}$ is divergence-free, and a supplied smooth tensor $S^{\lambda\mu\nu}$ satisfies

$$S^{\lambda\mu\nu} = -S^{\lambda\nu\mu}, \qquad \partial_\lambda S^{\lambda\mu\nu} = T^{\nu\mu} - T^{\mu\nu}.$$

This spin identity is a given hypothesis; it is not inferred from scalar Lecture 3. Define

$$B^{\lambda\mu\nu} = \frac{1}{2}\big(S^{\lambda\mu\nu} + S^{\mu\nu\lambda} + S^{\nu\mu\lambda}\big), \qquad \Theta^{\mu\nu} = T^{\mu\nu} + \partial_\lambda B^{\lambda\mu\nu}.$$

Prove that Θ is symmetric and divergence-free. Whenever the slice integrals exist, find the difference between its integrated four-momentum and that of T. Give a sufficient boundary condition for equality, and explain why local improvement alone is insufficient.

Hint. Check $B^{\lambda\mu\nu} = -B^{\mu\lambda\nu}$ first, and then compute $B^{\lambda\mu\nu} - B^{\lambda\nu\mu}$. For the charge, the time-derivative term has $B^{00\nu} = 0$.

Answer. The two algebraic identities are $B^{\lambda\mu\nu} = -B^{\mu\lambda\nu}$ and $B^{\lambda\mu\nu} - B^{\lambda\nu\mu} = S^{\lambda\mu\nu}$. The momentum difference is the spatial boundary integral of $B^{i0\nu}$. Compact spatial support, or periodic cancellation on a torus, makes it zero.

Exercise 3.4 (Maxwell Hilbert stress, after Lecture 9). Return to this problem after Lecture 9. Let $F_{\mu\nu} = -F_{\nu\mu}$ be held fixed with both indices down while a Lorentz metric is varied. Use the density and Hilbert convention

$$L = -\frac{1}{4} g^{\mu\rho} g^{\nu\sigma} F_{\mu\nu} F_{\rho\sigma}, \qquad \delta S = \frac{1}{2}\int T_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g, \quad k^{\mu\nu} = \delta g^{\mu\nu}.$$

The volume variation is $\delta \mathrm{vol}_g = -\frac{1}{2} g_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g$. Derive $T_{\mu\nu}$ by varying both inverse metrics. At $g = \eta$, set $F_{i0} = E_i$ and $F_{ij} = \epsilon_{ijk} B_k$, with $\epsilon_{123} = 1$. Compute T_{00} and T^{i0} in terms of the vectors $\mathbf{E}$ and $\mathbf{B}$. Evaluate them for $\mathbf{E} = (1, 2, 0)$ and $\mathbf{B} = (0, 0, 3)$.

Hint. Antisymmetry of F makes the two varied inverse metrics contribute equally. A lower spatial index introduces a minus sign when raised; distinguish T_{i0} from T^{i0}.

Answer. $T_{\mu\nu} = -F_{\mu\alpha} F_\nu{}^\alpha + \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}$. The energy density is $(|\mathbf{E}|^2 + |\mathbf{B}|^2)/2$, and $T^{i0} = (\mathbf{E} \times \mathbf{B})_i$. The example has energy density 7 and flux $(6, -3, 0)$.